MUHAMMAD FAIZAN KHAN

AI Engineer | Computer Vision & Generative AI/LLM Systems

Pakistan | +92 3174558424 | faizankhan58396@gmail.com
linkedin.com/in/muhamad-faizan-khan | github.com/faizankhan789



Professional Summary

AI Engineer with 3+ years of experience building production Large Language Model (LLM) and Vision-Language Model (VLM) pipelines, computer vision systems, and Azure AI workflows — cutting end-to-end pipeline runtime 30–50% through async parallelization. At NexaQuanta, built multi-provider LLM/VLM video-inspection pipelines, an Azure-based Retrieval-Augmented Generation (RAG) system, and a Stable Diffusion/ControlNet pipeline for photorealistic car imagery. Strong foundation in Python, deep learning, object detection/tracking, and MLOps.

Technical Skills

- **Programming Languages:** Python, C++, Swift, SQL, MySQL, HTML, CSS
- **AI/ML:** Large Language Models (LLM), Vision-Language Models (VLM), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Fine-Tuning (LoRA), Transformers, CNNs, RNNs, Object Detection (YOLOv5/v7/v8, Faster R-CNN, SSD), Object Tracking (Deep SORT, ByteTrack), Stable Diffusion & SDXL (ControlNet, ComfyUI), Optical Character Recognition (OCR)
- **Frameworks & Libraries:** TensorFlow, Keras, scikit-learn, PyTorch, OpenCV, LangChain, Flask
- **Data & Infrastructure:** Docker, Redis, PostgreSQL, AWS (S3, SQS), Azure AI (Document Intelligence), Apache Kafka, Vector Databases (Pinecone, ChromaDB, FAISS), FFmpeg, Git
- **Practices:** MLOps, Model Deployment, CI/CD, REST APIs, Async & Concurrent Programming, Hyperparameter Tuning, Cross-Validation, Model Evaluation, Natural Language Processing (NLP), Data Annotation (LabelImg, Roboflow)

Work Experience

AI Engineer *Jun 2025 – Present*

NexaQuanta

Design, train, and optimize deep learning models for real-world automation, collaborating with cross-functional engineering teams to build complete pipelines from data preprocessing through deployment across multiple client engagements.

Altia

- Developed a robust data extraction pipeline using Mistral OCR and Azure Document Intelligence to structure and process large volumes of unstructured documents for downstream AI workflows.
- Designed and implemented an agentic system with specialized agents to autonomously perform distinct tasks, enabling multi-step reasoning and more context-aware decision-making.
- Built an Azure-based Retrieval-Augmented Generation (RAG) system, integrating Azure AI services with vector search to deliver accurate, context-grounded responses at scale.

CitNow — Vehicle Inspection Video Highlight Generator

- Orchestrated an async, multi-provider LLM/VLM pipeline (OpenAI, Gemini multimodal, Mistral, YOLO, OpenCV, Redis, AWS S3, FFmpeg) with dynamic routing that fuses transcript analysis, video understanding, and object detection into a confidence-weighted highlight/timestamp engine with priced condition tags.
- Built a resilient JSON-schema LLM output layer with truncation repair, retries, and cost tracking; cut end-to-end pipeline runtime 30–50% via async parallelization.

CitNow — Video Quality Scoring Pipeline

- Built an async, multi-modal pipeline (AWS SQS/S3, Redis, PostgreSQL, ElevenLabs STT, Gemini/GPT-4.1, YOLO, Docker) that transcribes and auto-scores inspection videos against a configurable checklist using LLM (speech), VLM (visual), and YOLO (object-detection) checks, with retry/caching and webhook-driven delivery.
- Contributed an automated accuracy-testing suite benchmarking pipeline output against ground-truth data.

CitNow — Imaging Product

- Built an end-to-end realistic background generation pipeline for car imagery using Stable Diffusion/SDXL workflows with Tiling ControlNet, Depth ControlNet, BiRefNet, and IP-Adapter for high-fidelity composition and structural consistency.
- Applied a fine-tuned SDXL LoRA model for natural shadows, reflections, and realistic tinted windows; fine-tuned SDXL/Stable Diffusion models with Kohya SS for custom, product-tailored image generation.
- Designed a production-ready Streamlit app for automated logo placement on car images, including background analysis, placement logic, adaptive blending, and verification.
- Integrated trained models and ControlNet workflows into ComfyUI, enabling a modular, scalable pipeline for production-grade image generation.

AI & Machine Learning Engineer *Feb 2024 – May 2025*
Edraak Systems

- Developed ML models for operational efficiency across computer vision and NLP tasks — feature engineering, algorithm optimization, dataset preprocessing, training, and evaluation with advanced validation techniques.

Machine Learning Engineer *Jan 2023 – Feb 2024*
HiTech Provider

- Built ML models for workflow automation and accuracy improvements — preparing large datasets, experimenting with modern architectures, tuning hyperparameters, and evaluating with robust testing methodologies.

Machine Learning Engineer — Freelance *2022 – 2024*
Chegg

- Developed RAG systems for document summarization and context-aware QA using LangChain's Conversational Retrieval Chain with Pinecone for vector retrieval and multi-turn memory.
- Built a breast cancer classification model (Logistic Regression, SVM, Random Forest) with Python, Scikit-learn, and Seaborn for medical data analysis.
- Built a spam email detection model using Naive Bayes and SVM with Scikit-learn and NLTK for NLP text classification.

Projects

Saudi Number Plate Detection & Recognition System | *Python, YOLO, OpenCV, OCR* [GitHub]

- Real-time YOLO + OpenCV license plate detection with a custom OCR engine for Saudi plates; processes live video for traffic monitoring at 95%+ accuracy.

Jarvis 2025 — AI Voice Assistant | *Python, Speech Recognition, NLP* [GitHub]

- AI voice assistant controlling system operations (app launching, web search, file management) via natural language, with weather/news/info APIs at 90% command accuracy.

Face Emotion Detection System | *TensorFlow, Keras, OpenCV, CNN* [GitHub]

- CNN model for real-time facial emotion recognition across 7 classes on 35,000+ samples, reaching 92% validation accuracy with transfer learning and live webcam integration.

Image Similarity & Animal Classification | *TensorFlow, Keras, VGG16, Flask* [GitHub]

- VGG16 transfer-learning image similarity and classification system (95% accuracy) with a Flask REST API and web interface for cosine-similarity duplicate detection.

ML Car Price Prediction & WisePrice Platform | *Python, Scikit-learn, XGBoost, Flask* [GitHub]

- Car price prediction with multiple ML algorithms (Gradient Boosting, XGBoost, SVR, Lasso, Ridge), Bayesian tuning, and a full-stack "WisePrice" web app; 94% R^2 on test data.

Video Comparison & Analysis Tool | *Python, OpenCV, NumPy* [GitHub]

- Frame-by-frame video comparison using block-based pixel analysis for similarity scoring and difference detection, multi-threaded for quality assessment and forensic analysis.

Library Assistant Chatbot | *Python, Groq GenAI, NLP* [GitHub]

- AI conversational chatbot on Groq Generative AI for library management and book recommendations, integrated with a live database for context-aware, real-time responses.

RAG-based Enterprise Knowledge Assistant | *Python, LangChain, OpenAI API, ChromaDB* [GitHub]

- Production RAG system over 10K+ enterprise documents (PDF, DOCX, TXT) with chunking and vector embeddings; ChromaDB sub-second semantic search and LangChain-cited answers.

n8n Intelligent Workflow Automation Platform | *n8n, REST APIs, PostgreSQL, TensorFlow* [GitHub]

- AI-powered n8n workflows automating email/web-form/CRM sync and an ML reporting pipeline that fetches TensorFlow inference results, runs ETL, and sends automated notifications.

Dynamic Web Scraping Suite | *Python, BeautifulSoup, Selenium, Scrapy* [GitHub]

- Web scraping toolkit combining BeautifulSoup, Selenium, and Scrapy for static and JS-rendered sites; scraped 100K+ records for market research and price tracking.

Education

BS Computer Science *2023*
Qarshi University — CGPA: 3.76 / 94%

Certifications

- **Neural Networks and Deep Learning** — DeepLearning.AI, Coursera [Verify]
- **Introduction to Neural Networks** — Great Learning [Verify]
- **Machine Learning Modeling** — Great Learning [Verify]
- **Machine Learning Pipeline** — Great Learning [Verify]
- **OWASP Top 10 LLM 2025: AI Security Essentials** — Udemy [Verify]